Income Elasticity of Demand (YED)

Exam code: 9BS0

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Definition and calculation of income elasticity of demand

- **Changes in income** result in changes to the **demand** for products
  - Businesses are interested **in how much** the **quantity demanded (QD) will change** for different products
- The **income elasticity of demand (YED)** reveals how **responsive** the change in **QD** is to a change in **income**

Calculation

- YED can be calculated using the following formula:

$$\text{YED} = \frac{\text{\% change in quantity demanded}}{\text{\% change in income}}$$

Worked Example

An individual's income falls from £450 per week to £405 per week. As a result, their demand for takeaway meals falls from five per week to three per week.

Calculate the YED for takeaway meals.

[4]

Step 1: State the YED formula:

$$\text{YED} = \frac{\text{\% change in QD}}{\text{\% change in Y}}$$ [1]

Step 2: Calculate the % change in QD:

$$\text{\% change in QD} = \frac{5 - 3}{5} \times 100$$ [1]

$$= -40\%$$

Step 3: Calculate the % change in Y:

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

$$= -10\%$$

Step 4: Insert the above values into the YED formula:

$$\text{YED} = \frac{-40\%}{-10\%}$$ [1]

$$= 4$$

Interpretation of numerical YED values

- The YED value can be **positive or negative**, and the value is important in determining the type of good
  - A good with a **positive YED** value is considered to be a **normal good**
    - Normal goods can be classified as **necessities** or **luxuries**
  - A good with a **negative YED** value is considered to be an **inferior good**

Interpretation of the numerical values of YED

Numerical value	Type of good	Explanation
>1	**Luxury** • Examples include cars, smart watches, foreign holidays, cinema visits, jewellery and branded goods	• Demand rises when income rises, and demand falls when income falls • Demand is responsive to a change in income (income elastic)
0–1	**Necessity** • Examples include staple food items, such as bread, milk, eggs and potatoes, as well as fuel and toothpaste	• Demand is not very responsive to a change in income (income inelastic)
<0	**Inferior** • Examples include public transport, domestic holidays, canned foods and	• Demand rises when income falls (negative income elasticity) • Demand falls when income

The factors influencing YED

- YED is influenced by **many factors in an economy** that change the wages of workers
  - During a **recession,** wages sometimes fall and, more likely, do not rise significantly
    - Demand for **inferior goods** rises while demand for **luxury goods** falls
  - During a period of **economic growth** and rising wages, demand for luxury goods increases while demand for inferior goods decreases
  - Other influences on income include **minimum wage legislation**, **taxation** and **increased international trade**
- YED is also influenced by the **nature of the good**, as discussed above
  - Luxury or necessity (both are classified as normal goods)
  - **Inferior** or **normal** good

The significance of YED to businesses

- Understanding the YED is useful to businesses, as it can **help** them **plan their production and products**
  - Planning in this way will help them generate higher profits and have less exposure to downturns in the economy

Production planning

- A business needs to **plan how much it is going to produce**, which will help it **determine the number of resources**, such as raw materials and labour, **it will need**
- If a business can determine the YED for its products and can accurately predict changes in income, then it can **plan whether to increase or decrease production**
- It can help managers with financial planning
- Production planning is **easier when the YED is relatively inelastic**, as demand is likely to be more constant

Product planning

- The **economy goes through different stages over time**, from recession to recovery and growth, and so **incomes** will **fluctuate**
- This is known as the **business cycle**
- During a **recession**, producers of **inferior goods** will benefit from **higher demand** but will lose out when incomes rise and consumers return to normal goods
- Some businesses might have different products in their **product portfolio** to take account of this. For example:

"Essentials" range of products to appeal to more budget-conscious shoppers
- VW owns Skoda, Audi and Porsche, and it has different products within its range to appeal to different income groups

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